CHAPTER 12: SYMMETRY AND PATTERNS EXERCISE SOLUTIONS

Comprehensive Solutions and Complete Step-by-Step Answers for All Textbook Exercises

Exercise 12A Solutions (Page 173)

Q1. For the following figures, does the dotted line represent a line of symmetry? Write Yes or No.

- **a) Black rectangle with vertical dotted line in center:** **YES**. A vertical line dividing a rectangle exactly in half is a line of symmetry because the left half exactly overlaps the right half.
- **b) Square with asymmetric checkmark shape inside:** **NO**. Although the square itself is symmetric along its diagonal, the checkmark shape inside is asymmetric and will not align when folded.
- **c) Rhombus with horizontal dotted line through middle:** **YES**. A horizontal line passing through the opposite vertices of a rhombus is a line of symmetry.
- **d) Ellipse with off-center diagonal dotted line:** **NO**. An ellipse is only symmetric along its major and minor axes which pass through the exact center. This dotted line is off-center and near the right edge, so folding will not produce overlapping halves.

Q2. Draw as many lines of symmetry as possible for each figure.

- **a) Rounded Square:** **4 lines of symmetry** (1 horizontal, 1 vertical, 2 diagonal). It retains the same symmetry lines as a standard square.
- **b) Semicircle:** **1 line of symmetry** (vertical line passing through the highest point of the arc and the midpoint of the flat diameter).
- **c) Double-Headed Block Arrow (pointing left-right):** **2 lines of symmetry** (1 horizontal line running along the arrow's length, and 1 vertical line through the middle of the shaft).
- **d) Symmetric Pentagon/Isosceles Trapezoid-like polygon:** **1 line of symmetry** (vertical line passing through the center of the shape dividing it into identical left and right halves).

Q3. Draw a mirror image along the given line of symmetry to get a symmetrical figure (drawn on a grid).

- **a) Half of an Hourglass/Bowtie on left:** Draw an identical reflected shape on the right of the vertical line. The upper and lower diagonals should slope symmetrically outward, meeting in the middle to form a complete **hourglass / bowtie shape**.
- **b) Shape with two dots representing eyes:** The vertical line of symmetry is in the middle of a horizontal rounded rectangle. Ensure that the dot on the left is placed at the exact same horizontal distance from the symmetry line as the dot on the right, producing a symmetric pair of **eyes**.
- **c) Left half of an 'M' shape:** Mirror the shape to the right of the vertical line. The center point should touch the symmetry line, and the right diagonal and vertical outer wall will complete a perfect, symmetric **capital letter 'M'**.

Exercise 12B Solutions (Page 174)

Q1. Observe the pattern and draw the next figure.

- **a) Circle with moving red dot:** The dot moves 90 degrees clockwise at each step. (Left -> Top -> Right -> ...).
  Next figure (Step 4): Circle with the dot at the **bottom-middle**.
  Following figure (Step 5): Circle with the dot at the **left-middle**.
- **b) Square with moving black dot:** The dot moves counter-clockwise along the corners. (Top-Left -> Bottom-Left -> Bottom-Right -> ...).
  Next figure (Step 4): Square with the dot in the **top-right corner**.
  Following figure (Step 5): Square with the dot in the **top-left corner**.
- **c) Alternating shapes (Diamond and Circle):** The pattern is Diamond, Circle, Diamond, Circle...
  Next figure (Step 4): A **Circle**.
  Following figure (Step 5): A **Diamond**.
- **d) Squares with alternating diagonal lines:** The diagonals alternate orientation. (Bottom-Left to Top-Right -> Top-Left to Bottom-Right -> ...).
  Next figure (Step 4): A square with a diagonal line from **top-left to bottom-right**.

Q2. Fill in the blanks according to the given pattern.

- **a) 3, 6, 9, 12, __, __, __**
  Answer: **15, 18, 21** (Rule: Add 3 each time / multiples of 3).
- **b) 11, 22, 33, 44, __, __, __**
  Answer: **55, 66, 77** (Rule: Add 11 each time / multiples of 11).
- **c) 15, 13, 11, 9, __, __, __**
  Answer: **7, 5, 3** (Rule: Subtract 2 each time / decreasing odd numbers).
- **d) 70, 69, 68, 67, __, __, __**
  Answer: **66, 65, 64** (Rule: Subtract 1 each time / consecutive descending integers).
- **e) A26, B25, C24, __, __, __**
  Answer: **D23, E22, F21** (Rule: Letters go forward A to Z; numbers go backward 26 to 1).
- **f) ZAZ, YBY, XCX, __, __, __**
  Answer: **WDW, VEV, UFU** (Rule: First and third letter go backward; middle letter goes forward).
- **g) BA, DC, FE, __, __, __**
  Answer: **HG, JI, LK** (Rule: Consecutive letters grouped in pairs and written backward).
- **h) 1B, 2C, 3D, 4E, __, __, __**
  Answer: **5F, 6G, 7H** (Rule: Numbers increase by 1; letters go forward starting from B).

Exercise 12C Solutions (Pages 176 - 177)

Q1. Observe the patterns and fill in the blanks.

- **a) 72, 64, 56, 48, 40, __, __, __**
  Answer: **32, 24, 16** (Rule: Subtract 8 each time).
- **b) 965, 865, 765, 665, 565, __, __, __**
  Answer: **465, 365, 265** (Rule: Subtract 100 each time).
- **c) 4, 12, 36, 108, 324, __, __, __**
  Answer: **972, 2916, 8748** (Rule: Multiply each term by 3).

Q2. Fill in the blanks.

- **a) Sum of 4 Consecutive Numbers (increases by 4 each step):**
  1+2+3+4 = 10
  2+3+4+5 = 14
  3+4+5+6 = **18**
  4+5+6+7 = **22**
- **b) Sum of 5 Consecutive Numbers (increases by 5 each step):**
  1+2+3+4+5 = 15
  2+3+4+5+6 = 20
  3+4+5+6+7 = **25**
  4+5+6+7+8 = **30**
- **c) Sum of Even Numbers (increases sequentially by next even terms):**
  2+4 = 6
  2+4+6 = 12
  2+4+6+8 = 20
  2+4+6+8+10 = **30**
  2+4+6+8+10+12 = **42**
- **d) Consecutive Integer Sums:**
  1+2 = 3
  1+2+3 = 6
  1+2+3+4 = 10
  1+2+3+4+5 = **15**
  1+2+3+4+5+6 = **21**
- **e) Dividing 9s by 9 (Results in repeating 1s matching number of 9s):**
  (100 - 1) / 9 = 99 / 9 = 11
  (1000 - 1) / 9 = 999 / 9 = 111
  (10000 - 1) / 9 = **1111**
  (100000 - 1) / 9 = **11111**
  (**1000000 - 1**) / 9 = 111111
- **f) Multiplication patterns of 8 and 20:**
  160 / 8 = 20
  1600 / 8 = 200
  16000 / 8 = **2000**
  160000 / 8 = **20000**
  (**1600000**) / 8 = 200000

Exercise 12D Solutions (Page 178)

Q1. Using the given coding scheme, write these messages in secret code.

Cipher Mapping: A->m, B->n, C->o, D->p, E->q, F->r, G->s, H->t, I->u, J->v, K->w, L->x, M->y, N->z, O->a, P->b, Q->c, R->d, S->e, T->f, U->g, V->h, W->i, X->j, Y->k, Z->l.

- **a) MISSION TO MARS:**
  M(y) I(u) S(e) S(e) I(u) O(a) N(z) T(f) O(a) M(y) A(m) R(d) S(e)
  Secret Code: yeesuaz fa ydme (or **YEESUAZ FA YDME**)

- **b) TURN LEFT:**
  T(f) U(g) R(d) N(z) L(x) E(q) F(r) T(f)
  Secret Code: fgdz jqrf (or **FGDZ JQRF**)

Q2. Decode the secret messages using the same scheme.

- **a) tai mdq kag:**
  t(H) a(O) i(W) m(A) d(R) q(E) k(Y) a(O) g(U)
  Decoded Message: HOW ARE YOU

- **b) eqzp ftq yazqk:**
  e(S) q(E) z(N) p(D) f(T) t(H) q(E) y(M) a(O) z(N) q(E) w(K)
  Decoded Message: SEND THE MONEY

Exercise 12E Solutions (Page 178)

Q1. State whether the following shapes tessellate or not.

- **a) Parallelogram: YES**. Any parallelogram can repeat side-by-side to cover a flat plane with zero gaps or overlaps.
- **b) Trapezoid: YES**. An arrangement of normal and inverted trapezoids fits together perfectly without gaps.
- **c) Pacman-like shape (cut circle): NO**. The curved outer arcs and angular cuts cannot align repeatedly without leaving gaps.
- **d) Curved wave/interlocking shape: YES**. Because the convex curved side of one shape fits exactly into the concave side of the next, it tessellates perfectly like a jigsaw puzzle.

Workout Solutions (Pages 179 - 180)

Q1. Observe the pattern and draw the next figures in the empty boxes.

- **a) Dice dot series:** The number of dots increases by 1 at each step (1 -> 2 -> 3 -> 4).
  Next Box (Step 5): Draw **5 dots** (four corners + center).
  Final Box (Step 6): Draw **6 dots** (arranged in two vertical columns of three).
- **b) Rotating block arrow:** The arrow rotates 90 degrees clockwise at each step (Right -> Down -> Left -> Up).
  Next Box (Step 5): Arrow pointing **Right**.
  Final Box (Step 6): Arrow pointing **Down**.

Q2. Write the next three numbers according to the given pattern.

- **a) 20, 17, 14, 11, __, __, __: 8, 5, 2** (Rule: Subtract 3 each time).
- **b) 80, 70, 60, 50, __, __, __: 40, 30, 20** (Rule: Subtract 10 each time).
- **c) 1, 1, 2, 1, 1, __, __, __: 2, 1, 1** (Rule: Repeating block '1, 1, 2' repeatedly. Alternative is 3, 1, 1 if the third term increases).
- **d) 21, 28, 35, 42, __, __, __: 49, 56, 63** (Rule: Add 7 each time / multiples of 7).

- **e) 23, 26, 29, 32, __, __, __: 35, 38, 41** (Rule: Add 3 each time).
- **f) 108, 117, 126, 135, __, __, __: 144, 153, 162** (Rule: Add 9 each time).
- **g) 75, 66, 57, __, __, __: 48, 39, 30** (Rule: Subtract 9 each time).

Workout Solutions (Continued) & Special Challenges

Q3. Fill in the blanks.

- **a) Sum of consecutive odd numbers starting from different values:**
  1+3 = 4
  1+3+5 = 9
  1+3+5+7 = 16
  3+5+7+9 = 24
  5+7+9+11 = **32**
  7+9+11+13 = **40** (Rule: Sum increases by 8 at each step)

- **b) Number 9 expansion:**
  9+1 = 10
  99+1 = 100
  999+1 = **1000**
  9999+1 = **10000**

- **c) 104 * 4 series (inserts 0 in the middle of product):**
  104 × 4 = 416
  1004 × 4 = 4016
  10004 × 4 = 40016
  100004 × 4 = **400016**
  1000004 × 4 = **4000016**

- **d) Repeater 9s (reverses middle digits from 9 to 8 to 7):**
  111 × 9 = 999
  222 × 9 = 1998
  333 × 9 = 2997
  444 × 9 = **3996**
  555 × 9 = **4995**

Q4. State whether the following shapes tessellate or not (Page 180).

- **a) Green interlocking puzzle block: YES**. The indents and tabs lock together flawlessly with no gaps.
- **b) Blue D-shape / semicircle: NO**. Curved edges cannot fit together without gaps.
- **c) Red Cross (Greek cross): YES**. Crosses fit together snugly (alternating tabs and valleys) to completely tile a flat surface.

Q5. Secret message coding scheme (Page 180):

Key table: D->1, K->2, Q->3, Y->4, V->5, Z->6, C->7, F->8, J->9, R->a, M->b, P->c, U->d, L->e, A->f, W->g, I->h, S->i, X->j, G->k, T->l, E->m, N->n, O->o, B->p, H->q.

- **a) HAPPY BIRTHDAY:** H(q) A(f) P(c) P(c) Y(4) B(p) I(h) R(a) T(l) H(q) D(1) A(f) Y(4)
  Answer Code: qfcc4 phalq1f4

- **b) CYCLONE ALERT:** C(7) Y(4) C(7) L(e) O(o) N(n) E(m) A(f) L(e) E(m) R(a) T(l)
  Answer Code: 747eonm femal

- **c) TAKE COVER:** T(l) A(f) K(2) E(m) C(7) O(o) V(5) E(m) R(a)
  Answer Code: lf2m 7o5ma

- **d) WAIT TILL DAWN:** W(g) A(f) I(h) T(l) T(l) I(h) L(e) L(e) D(1) A(f) W(g) N(n)
  Answer Code: gfhl lhee 1fgn

Mental Maths & HOTS (Pages 180 - 181)

• **Mental Maths Q1. Capital letters looking identical in a mirror reflection:**
Of the options (P, W, M, L, O, R, X), the letters with vertical symmetry are: **W, M, O, X**.

• **HOTS Q1. Circle and dot sequence (Page 181):**
The dot alternates between the center and the edges. When at the edge, it moves clockwise (Right -> Bottom -> Left -> Top).
Box 8 (1st Blank): Circle with the dot at the **top inside edge**.
Box 9 (2nd Blank): Circle with the dot in the **exact center**.